

Dynamic probabilistic broadcasting in MANETs[☆]

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Abstract

In mobile ad hoc networks (MANETs), the efficiency of broadcasting protocol can dramatically affect the performance of the entire network. Appropriate use of a probabilistic method can reduce the number of rebroadcasting, therefore reduce the chance of contention and collision among neighboring nodes. A good probabilistic broadcast protocol can achieve higher throughput and lower energy consumption, without sacrificing the reachability or having any significant degradation. In this paper, we propose a probabilistic approach that dynamically adjusts the rebroadcasting probability as per the node distribution and node movement. This is done based on locally available information and without requiring any assistance of distance measurements or exact location determination devices. We evaluate the performance of our approach by comparing it with the AODV protocol (which is based on simple flooding) as well as a fixed probabilistic approach. Simulation results show our approach performs better than both simple flooding and fixed probabilistic schemes.

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1. Introduction

Network-wide broadcasting is used extensively in MANETs [1] for the process of route discovery, address resolution, and other network layer tasks. For example, on-demand routing protocols such as AODV [17] and DSR [11] use the broadcast information in route request packets to construct routing tables at every mobile node. The dynamic nature of MANETs, however, requires the routing protocols to refresh the routing tables frequently, which could generate a large number of broadcast packets at different nodes. Since every node in a MANET cannot directly communicate with the nodes outside its communication range, a broadcast packet may have to be rebroadcast several times at relaying

nodes in order to guarantee that the packet can reach all nodes. Consequently, an inefficient broadcast approach may generate many redundant rebroadcast packets.

The effectiveness of broadcasting is critical for the performance of MANET due to the following reasons. First, the wireless channel in a MANET is shared by neighboring nodes. Although CSMA/CA [10] algorithm is generally effective for collision avoidance, the contention among the broadcast/rebroadcast nodes for the channel will still substantially reduce the overall transmission rate. Moreover, wireless bandwidth is typically lower than its wired counterpart. Therefore, a poor broadcast approach can result in extremely low available bandwidth for data traffic.

There are many schemes proposed for broadcasting in MANETs. The simplest one is the flooding [15]. In this approach, as in wired network, each mobile host rebroadcasts received broadcast packets if they have not been received before. Packets that have already been recorded, are discarded. Since every mobile host rebroadcasts a packet for (and only for) the first-time it receives, it is easy to see that the total

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number of rebroadcasts is equal to $N - 1$, where N is the total number of mobile host in the MANET. In most cases, however, many such rebroadcast packets are redundant and simply waste the channel bandwidth.

More sophisticated solutions include probabilistic (gossip-based) [7,15,22], counter-based [15], distance-based [15,22], location-based [15,22], and neighbor-knowledge-based [22] approaches. In probabilistic approaches, a mobile host rebroadcasts packets according to a certain probability. In counter-based approaches, a packet will be rebroadcast only when the number of copies received at a host is less than a threshold. Distance- and location-based approaches further reduce redundant rebroadcasts by exploiting geographical information. The last two approaches require extra support from the physical layer or from the GPS devices. Neighbor-knowledge-based approaches explicitly exchange neighborhood information among mobile hosts. In this paper, we focus our study on the first two categories.

One problem of the probabilistic approaches is how to set the rebroadcast probability. Current approaches assume a fixed probability. It is demonstrated [7,15,22] that the optimal rebroadcast probability is around 0.65. Intuitively, this value is not likely to be globally optimal. For example, in a denser area, each mobile host has more neighbors whose coverage areas significantly overlap. Rebroadcast packets from hosts in a close neighborhood will reach the same subset of hosts many times. To reduce such redundancy, the rebroadcast probability in these areas should be set lower. On the contrary, the probability should be set higher in sparser areas so that a broadcast packet can reach all hosts in the MANET.

In this paper, we propose a dynamic probabilistic approach for broadcasting. We set the rebroadcast probability of a host according to the host density in its neighborhood area. When several hosts move toward each other to form a clique, their probabilities are set to be lower. When hosts move away from a dense area, their probabilities are kept higher. We use the packet counter in counter-based approaches to adjust the probability. If the packet counter is high, which means the host is in a dense area so that it can receive a large amount of rebroadcasts from its neighbors, we decrease the probability of this host. Otherwise, we increase the rebroadcast probability. We compare this approach with both the simple flooding approach and the fixed probabilistic approach by implementing them in a modified version of the AODV protocol. Simulation results show this simple adaptation can improve the average performance of broadcasting in various network scenarios.

The rest of this paper is organized as follows: In Section 2, we introduce the background and related work of broadcasting in MANETs. In Section 3, we describe our dynamic probabilistic approach, highlighting the difference in our approach from other similar approaches, and analyzing the strength and weakness of our approach. We analyze the lower bound on the number of retransmissions and time complexity in Section 4. We evaluate our approach and present

the simulation results in Section 5. Section 6 includes the concluding remarks of this paper.

2. Background and related work

Broadcasting in MANETs is an active research topic. The core problem is how to minimize the number of rebroadcast packets while maintaining good latency and reachability. Transmitting a large number of rebroadcasts does guarantee high reachability. However, it degrades the network throughput and potentially incurs long broadcast latency. On the contrary, dispatching fewer rebroadcasts leads to lower bandwidth wastage, higher throughput and lower broadcast latency. However, sending too few rebroadcasts may cause a rebroadcast chain broken so that some hosts may never receive the broadcast packet, thus resulting in low reachability. On the other hand, rebroadcast algorithms may also affect broadcast latency. For example, a host may dictate a random delay before sending a rebroadcast in order to reduce the chance of collisions. A long delay leads to high broadcast latency.

Whether or not to rebroadcast a packet at a certain host is determined by both the host's neighborhood and its running applications. If the network topology is known, and the application requires high reachability, the algorithm should take this as a dominant set problem [4,23,24] that is proved to be NP-complete. In reality, the network topology of a MANET is changing dynamically and usually unknown to mobile hosts. On the other hand, broadcasting is generally not required to be reliable so that reachability can be sacrificed in exchange for low rebroadcast overhead. Many approaches have been proposed for estimating neighborhood host density as well as conducting tradeoff between reachability and efficiency.

Broadcasting techniques can be categorized into the following five groups [15,22]: simple flooding, probabilistic, counter-based, area-based, and neighbor-knowledge-based. Other methods to relieve broadcast storm problem are also proposed in the literature, e.g. Hu et al. [9] introduced some schemes by using directional antennas. In the simplest approach of flooding, each mobile host rebroadcasts received broadcasting messages that are received for the first time, and records them for later references. Because every mobile host relays broadcasting messages just once, the total number of message relays is $N - 1$ if the network is not partitioned, where N is the total number of mobile host. In most cases, however, many such relays are redundant and waste the channel bandwidth. To reduce the number of redundant message relays, both probabilistic and deterministic methods are proposed.

Probabilistic methods such as gossiping have long been adopted to address the broadcast/multicast problem on wired networks [8,24]. Recent research work attempts to apply these methods to wireless networks. Krishnamachari et al. [13] analyzed the phase transition phenomena in wireless

ad hoc networks. Sasson et al. [18] applied random graphs [19] and percolation theory [20] to mobile ad hoc networks. They claim that there exists a threshold $P_c < 1$, such that by using P_c as the rebroadcast probability, almost all nodes can receive a broadcast packet, while there is no much improvement on reachability for $p > P_c$. Since P_c is different in various MANET topologies, and there is no existing method for estimating P_c , many probabilistic approaches use a predefined value for P_c . Let the rebroadcast probability be p . Assuming the total number of mobile hosts is N , the average total number of message relays will be $p * N$. The advantage of this method is its simplicity. However, not all routes are assessed by this approach. Routing protocols built on this approach may be unable to find the optimal route between a given source and destination node pair. On the other hand, an optimal probability for one topology may be suboptimal for other topologies. In this paper we attempt to address this problem by dynamically adjusting the rebroadcast probability.

Besides probabilistic methods, many deterministic approaches have also been studied. Ni et al. [15] introduced a counter-based approach. They analyzed the additional coverage of each rebroadcast after receiving n copies of the same packet. It shows, for $n = 1$, the maximum additional coverage is 61% of the original area, and the average additional coverage is 41%. The value decreases dramatically as n increases, which means the more copies a node receives, higher is the chance of its neighbors having already received the same packet, and the more likely is a rebroadcast redundant. In their approach, before transmitting a rebroadcast, a mobile node initiates a random assessment delay (RAD) and counts the number of received copies of the current packet. When the RAD expires, the node will rebroadcast the packet only if the counter does not exceed a threshold value C . Otherwise, the rebroadcast is dropped. The predefined threshold C is the key parameter in this approach. They showed many rebroadcasts could be saved when choosing C equal to 3 or 4. In their follow-on work [21], they showed that if choosing $C > 6$, few rebroadcasts can be saved in sparser networks. We use this result to set the threshold in our approach. This approach exhibits good performance, but it introduces packet delay at each hop. It is obvious that this approach is not suitable for delay-sensitive applications. In this paper we attempt to combine the counter-based approach with the probabilistic approach so that both low bandwidth consumption and low latency can be achieved simultaneously.

Ni et al. [15] also discussed area-based algorithms, including distance-based and location-based approaches. In distance-based approach, only neighbors far away from the current node rebroadcast packets. In location-based approach, each node receiving a packet compares its location with that of the previous-hop node, and calculates the additional coverage for the rebroadcast. If the value is below a threshold, the packet is dropped. Otherwise, the node starts a RAD. It recalculates the additional coverage whenever a new copy is received during the delay. It rebroadcasts the

packet if the value exceeds the threshold when the RAD expires. Otherwise the packet is dropped. The calculation of the additional coverage becomes complicated when several copies of the same packet are received. This kind of approach requires the assistance of location devices to provide location information of each mobile node.

Peng et al. [16] and Lim et al. [14] proposed two different neighbor-knowledge-based approaches. These approaches require mobile hosts to periodically exchange HELLO messages between neighbors. One such method, flooding with self-pruning, constructs a 1-hop neighbor list at each host from the HELLO messages. The neighbor list at the current host is added to every broadcast packet. When the packets reach the neighbors of the current host, each neighbor compares its neighbor list with the list recorded in the packets. It rebroadcasts the packets if not all of its own neighbors are included in the list recorded in the packets. Another approach of scalable broadcast algorithm (SBA) embeds neighbor list in HELLO messages from which it constructs a 2-hop neighbor list at each host. Neighbor-knowledge-based approaches make rebroadcast decisions based on the precise neighborhood information. Therefore, the number of rebroadcasts may be near optimal. However, the HELLO messages themselves consume channel bandwidth, thus affecting the overall performance.

Among the broadcasting techniques described above, the simplest one is flooding, which also generates the highest number of redundant rebroadcasts. The probabilistic approaches reduce the number of rebroadcasts at the expense of reachability. Counter-based approaches have better throughput and reachability, but suffering from relatively longer delay. Area-based algorithms require support from GPS or other location devices, and the neighbor-knowledge-based algorithms require the exchange of neighborhood information among hosts. Here, we propose an approach that combines the advantages of probabilistic and counter-based approaches to yield higher throughput, better reachability, and lower latency. Moreover, it is simple enough for easy implementation. We describe the details of our approach in the next section.

3. Dynamic probabilistic broadcasting algorithms

As described previously, the rebroadcast probability P is fixed in current probabilistic approaches. However, the threshold value P_c may be different for different topologies. For example, P_c of a sparse network is high while P_c of a dense network is relatively low. If P is set to a far smaller value than P_c , the reachability will be poor. On the other hand, if P is set far larger than P_c , many redundant rebroadcasts will be generated. In order to achieve both high throughput and high reachability in MANETs where network topology changes frequently, the rebroadcast probability at every host must be dynamically adjusted.

Intuitively, the rebroadcast probability should be set high at the hosts in sparser areas and low at the hosts in denser areas. One simple method for density estimation is to use packet counter. We maintain a counter at each host for every received broadcasting packet. The counter increases whenever a copy of the packet is received. A high counter value implies that the number of neighbors of the current host is high. On the contrary, a low counter value corresponds to a small number of neighbors. Therefore, we use packet counters as density estimates.

The value of a packet counter does not necessarily correspond to the exact number of neighbors from the current host, since some of its neighbors may have suppressed their rebroadcasts according to their local rebroadcast probability. However, it is proportional to the host density in the surrounding area for a given rebroadcast probability distribution among neighboring hosts. We increase the probability if the value of a packet counter is too low (or equivalently if the current node is located in a sparse neighborhood), which indirectly causes the probability at neighboring hosts to be incremented, and in turn increase the value of the counter for the next packet at current host. Similarly, we decrease the probability if the value of a packet counter is too high. This kind of adaptation causes a dynamic equilibrium between rebroadcast probabilities and packet counter values among neighboring hosts. Intuitively, the probabilities at the equilibrium states should lead to optimal solutions.

In reality, however, it is hard to reach equilibrium state since the mobile hosts may be constantly moving. The neighborhood of a host may have changed before the probability converges to an optimal value. This observation implies that we should adjust the probability as quickly as possible. On the other hand, the value of a packet counter reflects the number of neighbors only in a statistical sense, even if the network topology is fixed. Therefore, we should avoid drastic changes in the probability. Combining these two factors, we adopt a simple adaptation algorithm: rebroadcast a packet according to the current probability P if the packet is received for the first N_c times, where N_c is the threshold value to indicate whether enough copies of the broadcasting packet was received or not; decrease the probability P by a small constant d when an additional copy beyond N_c of an existing packet is received; increase P by another small constant d_1 if a host has not heard anything within a time interval t ; set a fixed upper bound P_u and lower bound P_l for P . The algorithm is shown below in detail.

3.1. Upon receiving a broadcast packet

If it is in the received packet list
 If packet counter $> N_c$
 $P = P - d$
 If $P < P_l$ Then $P = P_l$
 Increase the packet counter by 1

Else

Set the packet counter to 1
 Add the ID of this broadcast packet to the
 received packet list with an expiration time
 Rebroadcast the packet according to the probability P

For every time interval t

If no broadcast packet is received within t Then

If packet counter $< N_c$

$P = P + d_1$

If $P > P_u$ Then $P = P_u$

Remove the packet id from the
 received packet list

The choice of the value of t is critical in our algorithm. If we set t too low, we may check the counter too often and the packet counter may remain low. Therefore, the probability value may stay the same. On the other hand, if we set t too high, we may check the counter too less and the packet counter may exceed the threshold often and the rebroadcast probability may be set too low.

Our algorithm is a combination of probabilistic and counter-based approaches. It dynamically adjusts the rebroadcast probability P at each mobile host according to the value of the local packet counters. The value of P changes when the host moves to a different neighborhood. In a sparser area, the rebroadcast probability is larger and in denser area, the probability is lower. Compared with the probabilistic approach where P is fixed, our algorithm achieves higher throughput since the total number of rebroadcasts is reduced. On the other hand, the decision to rebroadcast is made immediately after receiving a packet in our algorithm without any delay. Therefore the broadcast latency of our algorithm is lower than that of the counter-based approach where the rebroadcast packets are transmitted after a random delay.

3.1.1. Selection of initial value of P

The most important factor in our algorithm is the selection of the initial value of P . A larger P incurs more redundant rebroadcast while a smaller P leads to lower reachability. Even if our algorithm adjusts the rebroadcasting probability dynamically, it takes some time to reach the ideal value. So if the initial value is closer to the ideal value, our algorithm will adjust the rebroadcasting probability to the ideal value soon. In this section, we present the analysis of average neighbor number to provide the basis for the selection of the initial value of P .

Let A be the area of an ad hoc network, N be the number of mobile hosts in the network, and R be the communication range. The average number of neighbor N_{neighbor} can be obtained by the following formula:

$$N_{\text{neighbor}} = (N - 1) \frac{\pi R^2}{A}. \quad (1)$$

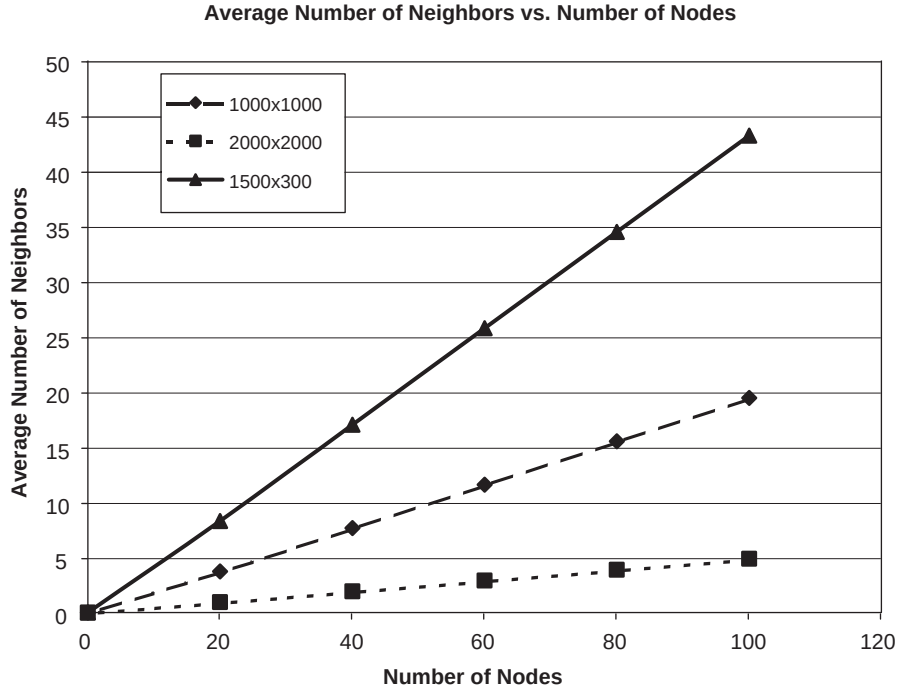


Fig. 1. Average number of neighbors vs. number of nodes.

Fig. 1 shows the average number of neighbors for different node number with the network area of 1000 m × 1000 m, 2000 m × 2000 m, and 1500 m × 300 m, respectively. The higher is the average number of neighbors, the denser the network is. Lower the average number of neighbors is, sparser the network is. From the average number of neighbors, we can calculate the initial value of N_{neighbor} by applying the result in [21], which shows if we choose $C > 6$, few rebroadcasts can be saved in sparser networks. So the initial value P can be calculated as follows:

$$P = \begin{cases} 1, & \text{where } \frac{6}{N_{\text{neighbor}}} \geq 1, \\ \frac{6}{N_{\text{neighbor}}}, & \text{where } 0 < \frac{6}{N_{\text{neighbor}}} < 1, \\ 0, & \text{where } \frac{6}{N_{\text{neighbor}}} \leq 0. \end{cases} \quad (2)$$

Fig. 2 shows the calculated initial value of P for different number of nodes with the network area of 1000 m × 1000 m and 1500 m × 300 m, respectively.

In next section, we will discuss the lower bound of the number of retransmission and time complexity.

4. Lower bound analysis

We analyze the lower bound on the number of retransmissions and time complexity in this section. In [6], a broadcasting algorithm based on broadcasting tree is introduced and the authors provide lower bound analysis on latency and total number of retransmissions. Both the latency and number of retransmission bounds are $O(1)$ times the optimal value for bounded transmission range in [6]. Several other papers introduced broadcasting algorithms and lower bound anal-

ysis on time complexity. For example, $O(D \log^2 n)$ in [5], $O(D 2^h \log^h n)$ in [3], $O(D \log(n/D) + \log^2 n)$ in [12] and $O(D + \log^5 n)$ in [2].

For simple flooding, the number of retransmission is $n - 1$. For probabilistic broadcasting, the number of retransmission is $(n - 1)p$. For our dynamic probabilistic scheme, the number of retransmission is greater than or equal to $(n - 1)P_l$ and less than or equal to $(n - 1)P_u$. Therefore, all three schemes have the complexity of $O(n)$ on the number of retransmissions.

Next, we analyze the time complexity of flooding, probabilistic broadcasting and our dynamic probabilistic scheme. We assume that all transmissions occur in slotted time as it makes it easier to analyze and this assumption holds if the broadcasting message is large enough. Let V be the whole set of mobile nodes in the network, i.e. $V = \{v_0, v_1, \dots, v_{n-1}\}$ where v_i denotes the mobile node. We assume v_0 is the source node. Let R_i be the set of nodes that received the broadcasting message after i step and T_i be the set of nodes that already received the broadcasting message and is ready to retransmit it after i step. Note that T_i does not include the nodes that already retransmit the message. We let T'_i be the set of nodes that actually retransmit the message at i th step. Therefore, we have $R_0 = \{v_0\}$, $T_0 = \{v_0\}$ and $T'_0 = \{v_0\}$. We use $N(X)$ denote the set of nodes which is neighbor node of any node in X .

From i th step to $i+1$ th step, some of the nodes in T_i retransmit the message. So we have

$$R_{i+1} = R_i \cup N(T'_i), \quad (3)$$

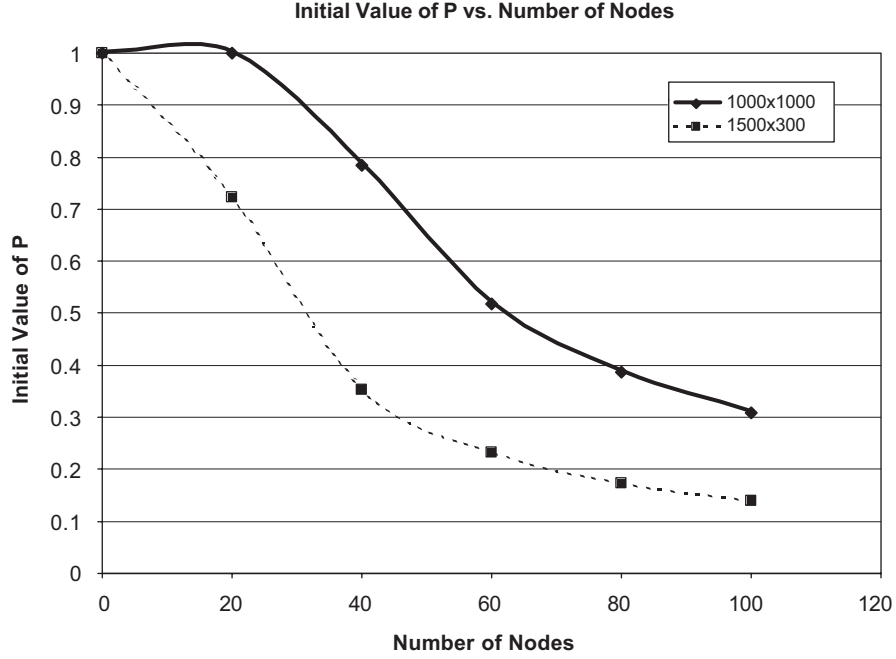
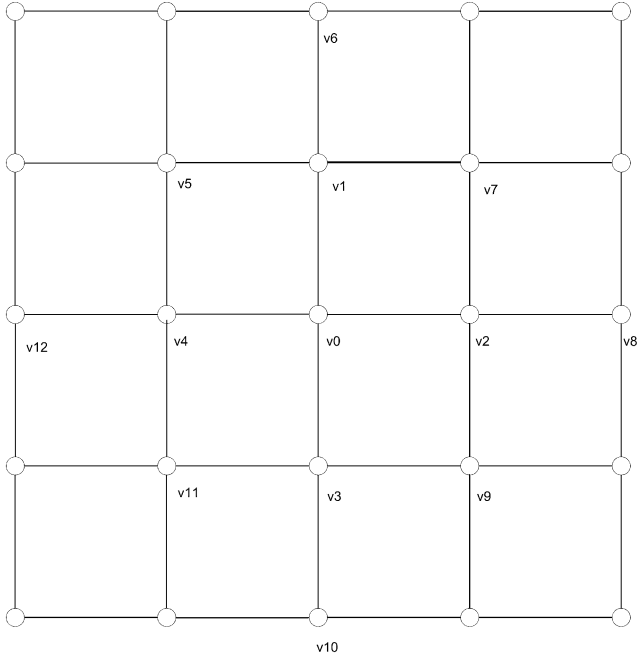
Fig. 2. Initial value of P vs. number of nodes.

Fig. 3. Grid layout.

$$T_{i+1} = R_{i+1} - \bigcup_{k=0}^i T'_k, \quad (4)$$

T'_{i+1} is a subset of T_i and all nodes in T'_{i+1} are not in the range of other nodes in T'_{i+1} .

The total time until all nodes get the broadcasting message depends on the choice of T'_{i+1} in each step. First, we consider

grid layout as shown in Fig. 3.

$$R_0 = \{v_0\}, \quad T_0 = \{v_0\} \quad \text{and} \quad T'_0 = \{v_0\}.$$

$$R_1 = \{v_0, v_1, v_2, v_3, v_4\}, \quad T_1 = \{v_1, v_2, v_3, v_4\} \quad \text{and} \\ T'_1 = \{v_1, v_2, v_3, v_4\}.$$

$$R_2 = \{v_0, v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8, v_9, v_{10}, v_{11}, v_{12}\}, \\ T_2 = \{v_5, v_6, v_7, v_8, v_9, v_{10}, v_{11}, v_{12}\} \quad \text{and} \\ T'_2 = \{v_5, v_6, v_7, v_8, v_9, v_{10}, v_{11}, v_{12}\}.$$

...

until $R_k = V$, $T_k = \emptyset$ and $T'_k = \emptyset$ for some $k > 0$.

So we have

$$|T_{i+1}| = 2|T_i| \quad \text{and} \quad |T_1| = 4, \quad (5)$$

$$N(T'_i) = T_{i+1}, \quad (6)$$

$|R_{i+1}| = |R_i| + |T_{i+1}| = |R_i| + 2^{i+2}$ then we have $|R_i| = 2 + \sum_{k=1}^i 2^k = 2^{i+1} + 1$.

It will take $\log_2(n-1)$ step to let all nodes receive the message for grid layout.

Generally, the choice of T'_{i+1} decides the total steps to have all nodes receive the broadcasting message. The size of T'_{i+1} depends on the size of T_i and the node placement in T_i . If the nodes in T_i are close to each other, less nodes can be selected in T'_{i+1} . More number of neighbor leads to more acceptance of the broadcasting message at each step, but the size of the candidate node that can retransmit the message without collision is still limited because the nodes may within the transmission range of each other.

Table 1
Simulation parameters

Simulation parameter	Value
Simulator	GloMoSim (v2.03)
Network range	1000 m × 1000 m
Transmission range	250 m
Number of mobile nodes	20, 40, 60, 80 and 100
Bandwidth	2 Mbps
Traffic type	Constant bit rate (CBR)
Packet rate	10 packets per second
Packet size	512 bytes
Simulation time	90 s
Number of trials	10

In each step, we choose at least one node from T_i to generate T'_i . So we have $\beta = |T'_i|/|T_i|$ and β has the range of $(1/n, 1)$ or $(1/x, 1)$ where x is the maximum number of neighbors of all nodes. Therefore, we can conclude that the flooding can be finished in $O(\log n)$. Moreover, because the probabilistic broadcasting and dynamic probabilistic scheme are similar in nature as flooding, they all have lower bound of $O(\log n)$ for time complexity.

5. Performance evaluation

In this section, we evaluate the performance of our dynamic probabilistic broadcasting algorithm. We compare our algorithm with a simple flooding algorithm and a probabilistic algorithm with fixed rebroadcast probability. We implement all the three algorithms in the AODV protocol. The metrics for comparison include the average number of routing request rebroadcasts, reachability, the average number of collisions, end-to-end delay, and the throughput.

5.1. Simulation setup

We use the GloMoSim network simulator [25] (version 2.03) to evaluate our algorithms. We study the performance of the broadcasting approaches in the context of higher-level application, namely, the AODV routing protocol [17] that is included in the GloMoSim package. The original AODV protocol uses simple flooding to broadcast routing requests. We have implemented two AODV variations: one using probabilistic method with fixed probability, called FP-AODV (AODV + fixed probability), and the other using our dynamic probabilistic algorithm, called DP-AODV (AODV + dynamic probability).

In our simulation, we use a 1000 m × 1000 m area with 20, 40, 60, 80 and 100 mobile hosts. The radio frequency at the physical layer is 2.4 GHz of the ISM band. The raw network bandwidth is 2 Mbps and the MAC layer protocol is IEEE 802.11 [10]. Other simulation parameters are shown in Table 1.

5.2. Analysis

The objective of this paper is to present the performance of a dynamic probabilistic broadcasting algorithms, using AODV and fixed probabilistic broadcasting approach as references. Our main idea is to reduce the rebroadcasting number in route discovery phase, therefore reduce the network traffic and decrease the probability of channel contention and packet collision. As a result, end-to-end delay can also be reduced and the throughput can be improved.

Since our algorithm is based on a probabilistic approach, it does not fit in every case. There is a small chance that the route requests cannot reach the destination in our probabilistic algorithm. In such cases, we have to generate the route request again if the previous route request failed to reach the destination. The AODV protocol, on the contrary, uses flooding in route discovery phase. Therefore, all route requests will reach their destinations if the network is not partitioned. Based on this analysis, our algorithm should perform better than AODV in dense networks.

5.3. Rebroadcasts

In AODV, a mobile node rebroadcasts all routing request packets that are received for the first time. Therefore, there are $N - 1$ possible rebroadcasts, where N is the total number of nodes. In FP-AODV, each node decides to rebroadcast or not, according to a fixed probability P . Since their decisions are independent, the total number of rebroadcasts is $P^*(N - 1)$ on the average. In DP-AODV, the rebroadcast probability is dynamically set. In sparser area, the probability is high and in denser area the probability is low. On the same network topology, the rebroadcast probability P in FP-AODV should be no less than the probability of DP-AODV nodes in sparse area in order to maintain the same level of reachability. As a result, the number of rebroadcasts in DP-AODV is definitely smaller than that of FP-AODV.

Next we compare the number of rebroadcasts in AODV, FP-AODV and DP-AODV through simulation. Fig. 4 shows that our algorithms can significantly reduce the number of rebroadcasts for a network with 100 nodes and maximum speed 10 m/s. It also shows the number of route request rebroadcast increases when the traffic load increases. The DP-AODV has the least number of rebroadcast for almost all traffic loads. We vary the traffic load by using different number of CBR source–destination connections. As shown in the figure, the saving is higher when the traffic load is heavier.

Fig. 5 shows the relationship between the number of relays and the mobility of a network with 100 nodes with 10 source–destination pairs. After we introduce mobility, more route requests are generated and some of them may fail to reach their destinations. Such failures cause another round of transmission of route request packets. Fig. 5 shows the number of route request rebroadcasts increases when the

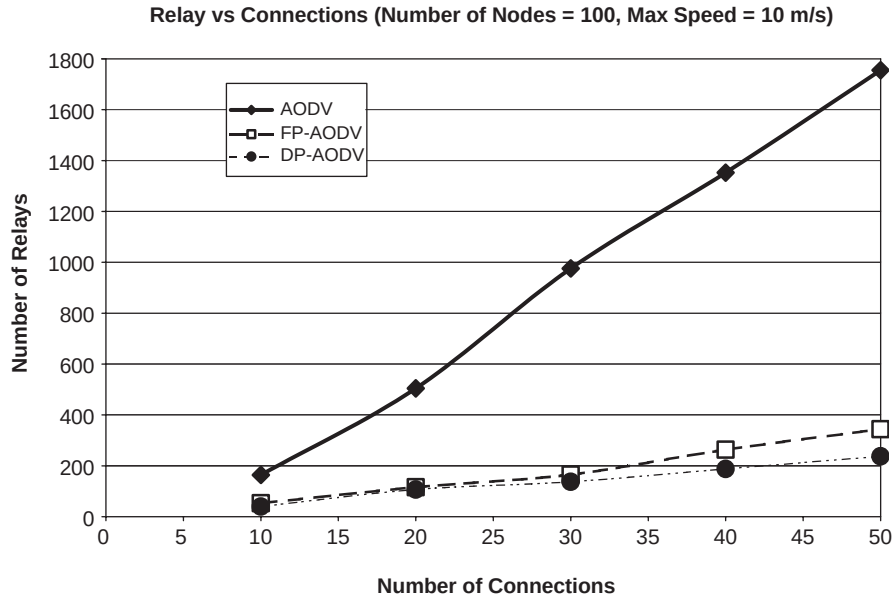


Fig. 4. The number of route request rebroadcasts vs. traffic load.

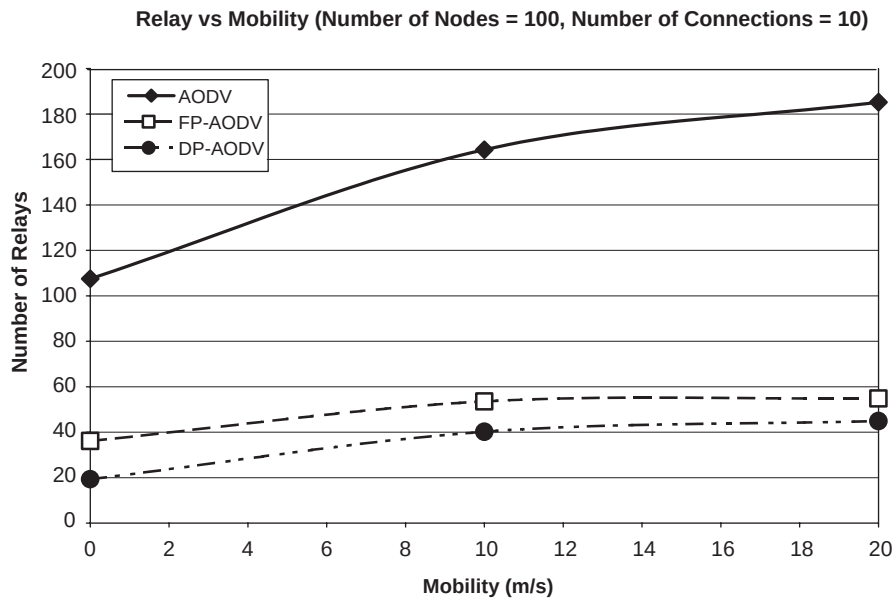


Fig. 5. The number of route request rebroadcasts vs. mobility.

mobility increases, and our approach has lower rebroadcast numbers than that of AODV and FP-AODV.

As shown in Figs. 4 and 5, FP-AODV can substantially reduce rebroadcast number in AODV. Using smaller forwarding probability can drastically reduce the number of route request rebroadcasts. However, sending too few route request rebroadcasts may result in request dying out, which incurs another round of route request. By using the method introduced in Section 3.1.1, we can obtain a proper initial value of rebroadcast probability.

5.4. Reachability

The metric of reachability measures the ratio of nodes that can receive a broadcast packet. A mobile host will miss a packet if all of its neighbors decide to suppress rebroadcasts. In a network without partition, the flooding approach guarantees that all nodes can receive the broadcast packets at the expense of extra cost caused by redundant rebroadcasts. In reality, however, redundant rebroadcasts also contribute to chances of packet collisions that may eventually

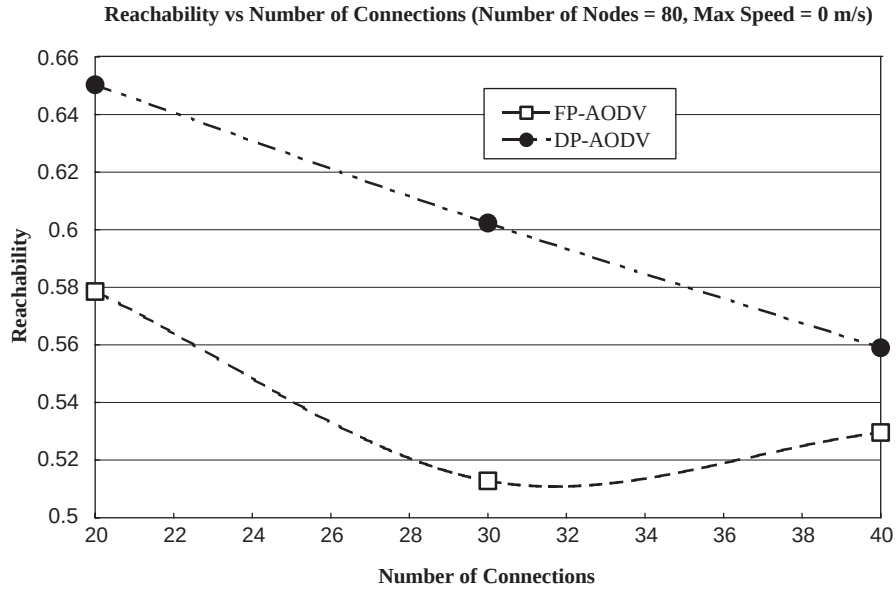


Fig. 6. Reachability vs. traffic load.

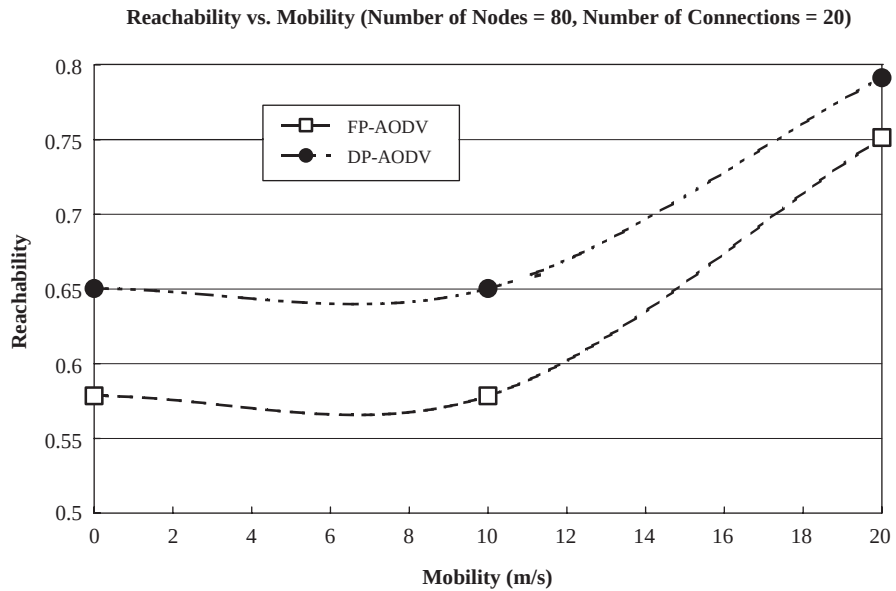


Fig. 7. Reachability vs. mobility.

cause packet drops, thus adversely affecting the reachability. Depending on the value of the probability, probabilistic approaches may have lower reachability compare to flooding-based approaches. However, by choosing appropriate probability value, we can achieve acceptable reachability. Moreover, the DP-AODV algorithm dynamically adjusts the rebroadcast probability so that it can achieve higher reachability than FP-AODV does.

We study the reachability in the context of the AODV routing protocol. We randomly select source–destination node pairs and check if a packet can reach the destination node from the source node. If there exists a route from the source

node to the destination node, then the routing request packets broadcast from the source node have reached the destination node. We calculate the ratio of the node pairs that have a route between the source and the destination over the total number of selected pairs. This ratio is not exactly equal to the reachability, but it is proportional to the reachability. We use this ratio to compare reachability of different approaches.

Fig. 6 shows the reachability for a network with 80 nodes, no mobility, and varying number of connections. Fig. 7 shows the reachability for a network with 80 nodes, 20 connections of source–destination pairs. The figures show that

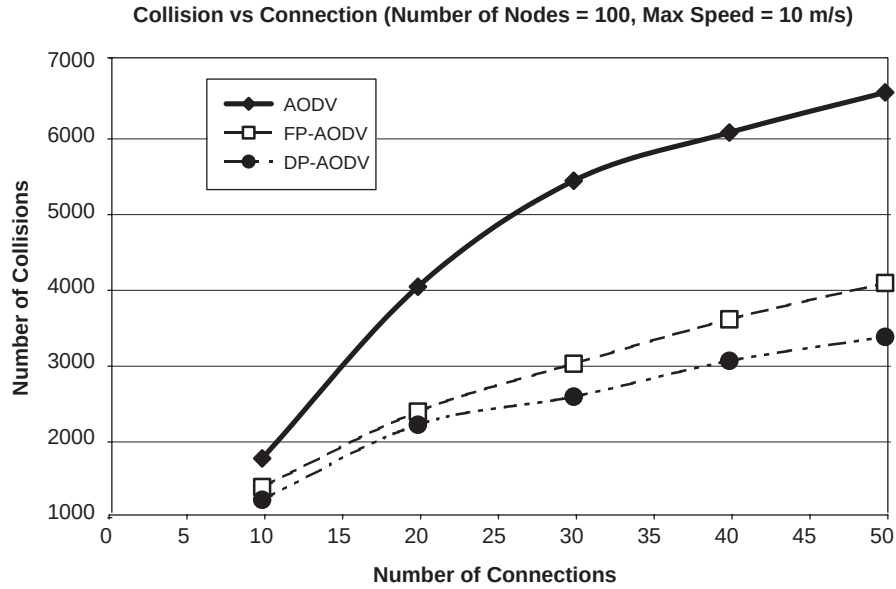


Fig. 8. The number of packet collisions vs. traffic load.

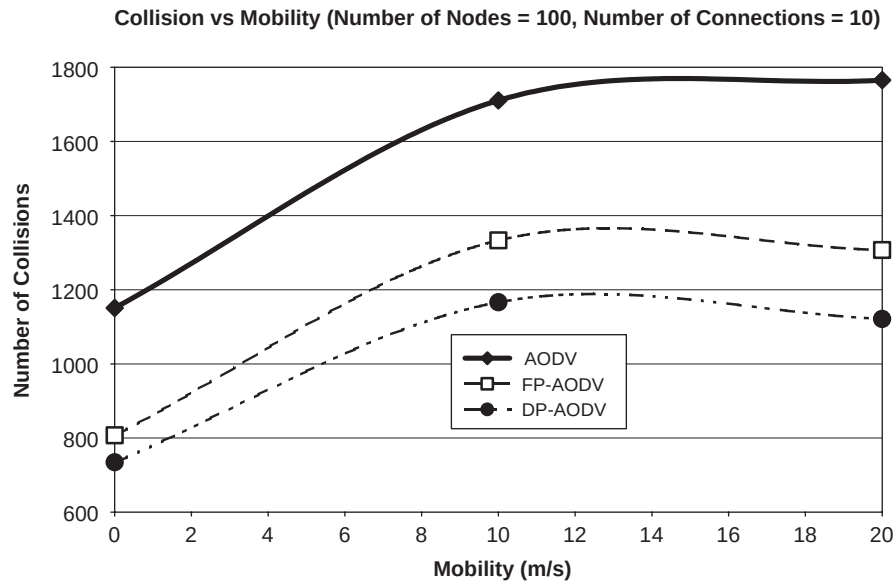


Fig. 9. The number of collisions vs. mobility.

DP-AODV has higher reachability than FP-AODV. It also implies that the dynamic probabilistic approach can achieve high throughput without sacrificing reachability.

5.5. Collisions

We measure the number of collisions for these schemes at the physical layer. Since data packets and control packets share the same physical channel, the collision probability is high when there are a large number of control packets. Fig. 8 shows the average number of collisions during the whole simulation period. The number of nodes is 100 and the maximum speed is 10 m/s.

As shown in Fig. 8, our algorithms incur fewer collisions. In Fig. 9, we show the performance with different mobility settings. When node mobility increases, more route requests fail to reach the destinations. In these cases, more route requests are generated, which lead to more collision due to the increase of control packets.

5.6. Latency

We measure the broadcast latency for the three approaches. We record the start time of a broadcast as well as the time when the broadcast packet reaches the last node.

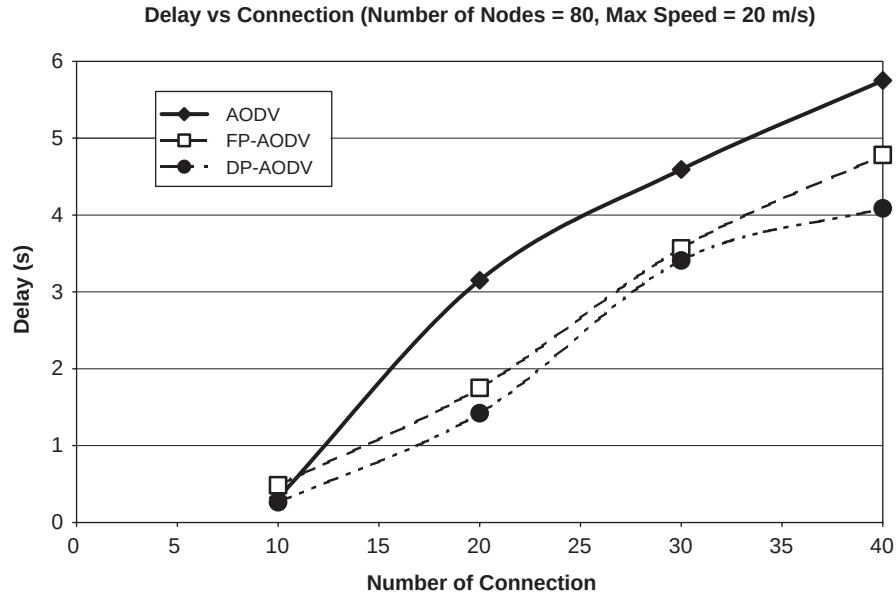


Fig. 10. Latency vs. traffic load (number of node = 80, max speed = 20 m/s).

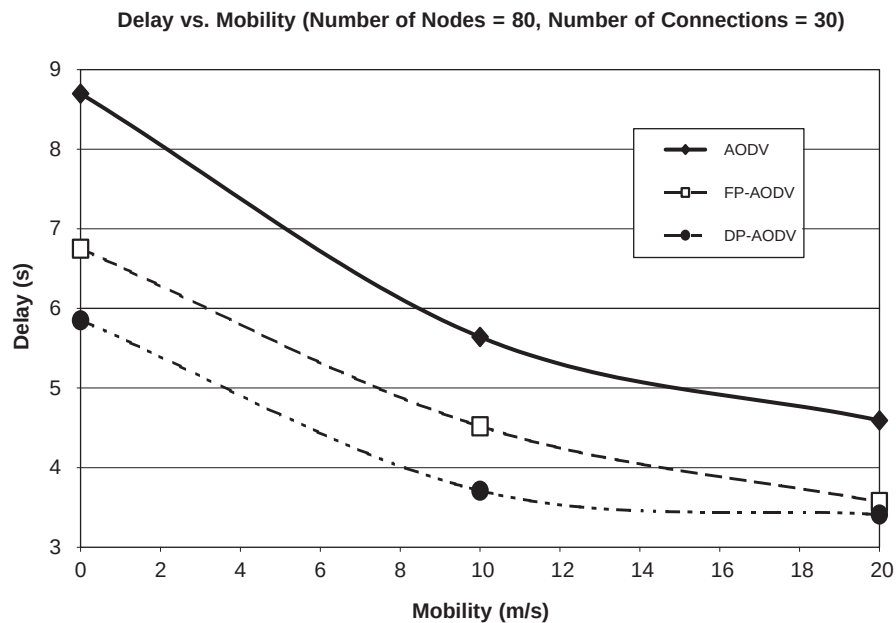


Fig. 11. Latency vs. mobility.

The difference between these two values is used as the broadcast latency. Since rebroadcasts collide and content for channel with each other, and the dynamic probabilistic approach incurs the lowest number of rebroadcasts, it should have the lowest latency.

The number of total packets transmitted in the channel has a significant impact on the latency. If the number of packets is high, then the number of collisions is also high, and in turn more retransmissions are needed. As a result, fewer packets lead to lower delays. Fig. 10 shows the latency for different levels of traffic load. As expected, our DP-AODV algorithms exhibit lower latency than the AODV and FP-AODV.

In Fig. 11, we compare latency for different mobility settings. The figure shows DP-AODV incurs lower latency. Therefore, for delay-sensitive applications, our scheme can definitely perform better.

5.7. Throughput

Throughput is an important metric that measures the transmission ability of a network. It is defined as the number of bits transmitted within a unit time. We compare the throughput in this section.

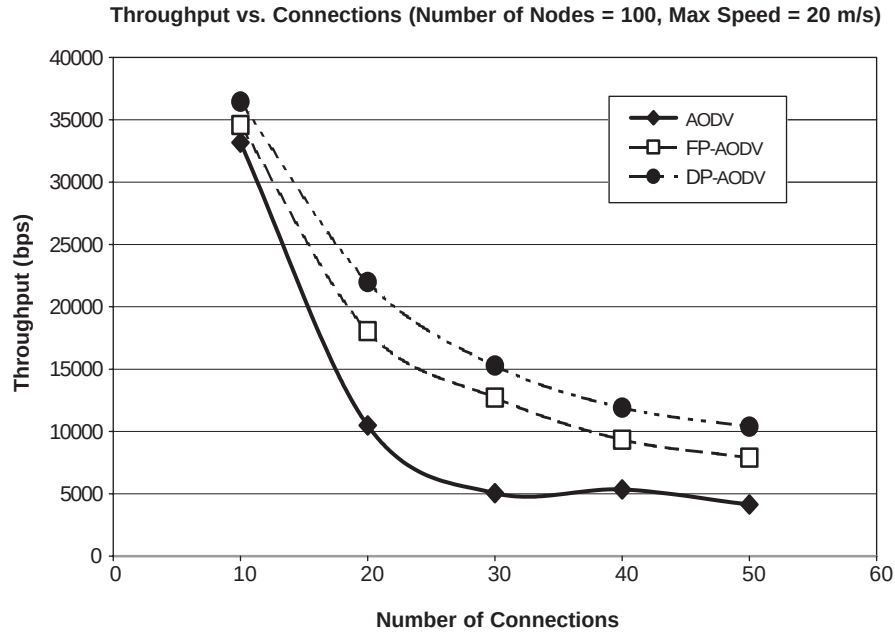


Fig. 12. Throughput vs. traffic load.

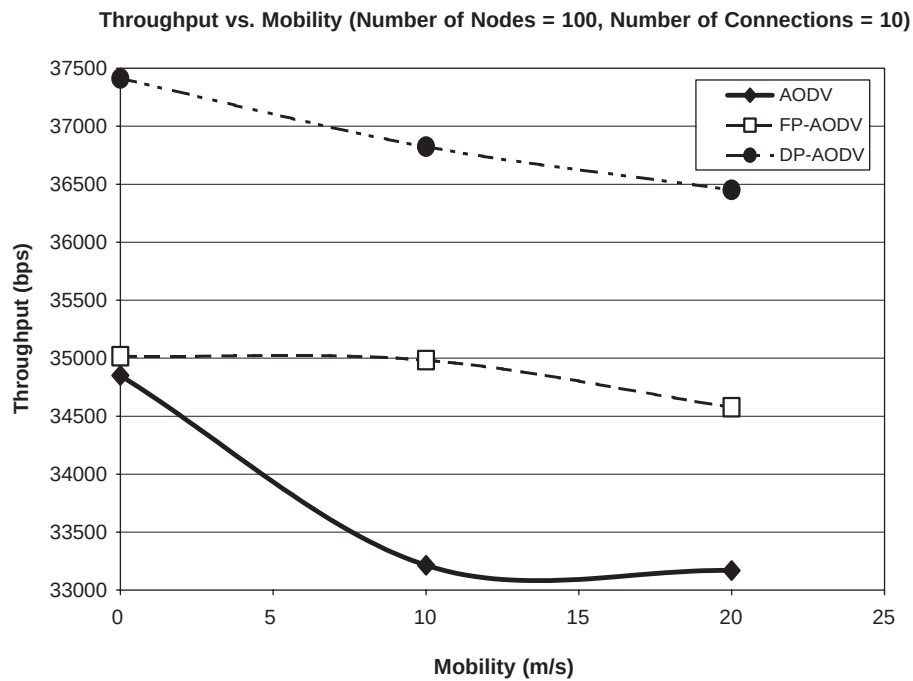


Fig. 13. Throughput vs. mobility.

Fig. 12 shows throughput decreases when traffic load increases. The more connections, the more route requests will be, leading to more rebroadcasts and higher bandwidth consumption and lower throughput.

In Fig. 13, we compare the network throughput for different mobility settings. It shows throughput decreases when mobility increases. The drop of throughput is caused by the fact that the movement of destination or intermediate nodes

may incur route expiration and route request retransmission. The figure also shows that DP-AODV outperforms AODV and FP-AODV. As mentioned before, the throughput improvement of DP-AODV is due to its reduction of rebroadcasting. The fewer rebroadcasts, the smaller bandwidth consumption by control messages is. This also results in lower degree of contentions and collisions, which leads to relatively higher throughput.

6. Conclusion

In this paper we introduce a dynamic probabilistic broadcasting approach for mobile ad hoc networks. Our scheme combines the fixed-value probabilistic approach with the counter-based approach. It dynamically adjusts the value of the rebroadcast probability at a mobile host according to the host density in its neighborhood. It uses the value of the packet counter to infer the host density. Our simulation results show this approach can generate fewer rebroadcasts than that of the fixed-value probabilistic approach. It also incurs lower broadcast latency than the counter-based approach does.

In our future work, we plan to build an analytic model for our dynamic probabilistic approach in order to facilitate the exploration of the optimal adaptation strategy. Based on the analytic model, we can obtain the proper value of the parameters in our approach and predict its performance in MANETs.

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